

Business Requirements & Project Management Assessment

Comprehensive Analysis, Gap Assessment, Tech Feasibility & Methodology Framework

Project Title:	Customer Data Analytics & Automated Workflow System	Date:	August 2026
Prepared By:	Lead Project Manager / Business Analyst	Version:	1.0 (Final)

1. Executive Summary & Business Needs Assessment

This document details the business requirements analysis and project management strategy for transitioning an organization from manual spreadsheets to an automated Python-based data analytics and communication platform.

- **Current State (AS-IS):** Customer activity, data analytics, and follow-ups are currently performed manually using basic spreadsheets. This process is time-consuming, error-prone, and lacks scalability.
- **Desired State (TO-BE):** A fully automated Python-driven system that provides real-time customer analytics, identifies popular product trends, and generates automated personalized outreach.
- **Gap Analysis & Business Drivers:**
 - *Efficiency Gap:* Manual data tracking vs. automated execution (Targeting a 70%+ reduction in processing time).
 - *Accuracy Gap:* Eliminating human errors in manual entry through automated script and API validation.

2. Requirement Elicitation & Active Listening Strategy

To ensure complete alignment with business goals and avoid ambiguous specifications, the following active listening and requirement gathering techniques were employed:

2.1 Elicitation Methodologies

- **Stakeholder Interviews:** Conducted one-on-one structured interviews with key operations leads to uncover underlying operational pain points and functional expectations.
- **Prototyping & Iterative Demos:** Developing early visual wireframes and functional working prototypes to collect real-time feedback and validate requirements.

2.2 Clear & Unambiguous Requirement Statements

Requirement Category	Requirement Statement (SRS Standard)
Functional (Data Input)	The system shall allow authorized users to record and store customer details, including full name, company name, phone number, email address, and mailing address.
Functional (Automation)	The system shall automatically execute daily data collection scripts and provide logs on performance metrics and completion status.

Requirement Category	Requirement Statement (SRS Standard)
Non-Functional (Performance)	The system API response time must remain under 2.0 seconds for processing real-time analytics queries under standard load.

3. Technical Feasibility Assessment

A thorough technical feasibility study was conducted to confirm that the proposed solution can be built reliably using modern software engineering tools and infrastructure.

Evaluation Parameter	Assessment Findings & Technical Strategy
Tech Stack Availability	Python 3.11+, FastAPI (REST APIs), Scikit-learn (Machine Learning), Prometheus & Grafana (Monitoring & Metrics).
System Scalability	Architecture must support expected user growth, high concurrency, and distributed logging/tracing for fault isolation.
Data Integration	Feasible data migration paths established to transform unstructured spreadsheet data into structured relational schemas.
Team Capabilities	Development team possesses strong Python and API expertise; targeted upskilling scheduled for advanced ML pipeline deployment.

4. Methodology Comparative Analysis: Agile vs. Waterfall

Selecting the appropriate project management framework is essential for managing delivery risks and accommodating evolving requirements.

Project Feature	Agile Framework (Selected Approach)	Traditional Waterfall Model
Process Flow	Iterative and Adaptive (2-Week Sprints)	Linear and Sequential (Strict Phase Gates)
Requirements Management	Flexible; welcomes changing business needs mid-project	Rigid; requires complete upfront freeze
Client Collaboration	Continuous feedback during Sprint Reviews & Demos	Limited interactions after initial requirements phase
Risk & Quality Control	Continuous integration & automated testing per Sprint	Late-stage testing; defect discovery occurs near release

Methodology Justification: Agile was selected because the client's business needs require ongoing exploration of features like personalized recommendations and email automation. The iterative loop allows

the team to deliver early working software increments and adapt rapidly based on continuous stakeholder feedback.

5. Key Performance Indicators (KPIs) & Success Metrics

To measure the success of the technical solution against core business needs, the following KPIs will be tracked:

- Processing Time Reduction:** Execution time comparison between the automated script and the original manual spreadsheet task.
- System Reliability & Accuracy:** Achieving 99.5% script execution uptime with zero critical bugs reaching production.
- Delivery Velocity:** On-time completion of bi-weekly Sprint releases with fully tested, client-approved increments.

6. Document Approval & Sign-off

Lead Project Manager

Lead Technical Architect

Client / Sponsor